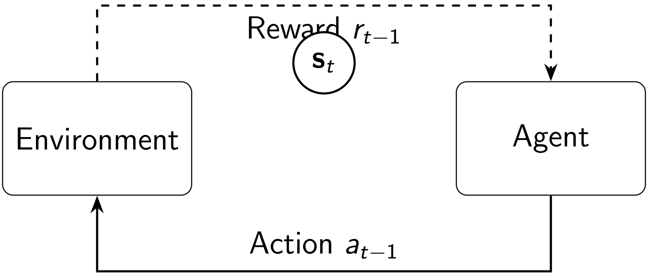
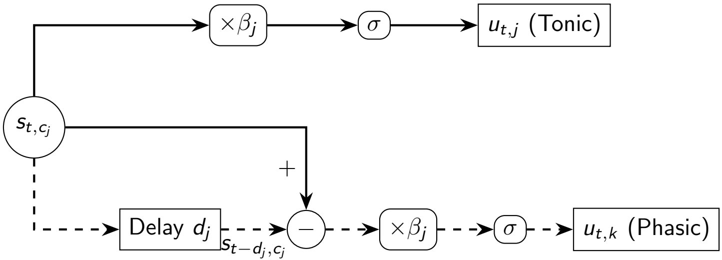
Equation 1

$$\mathbf{s}_t = \begin{bmatrix} a_{t-1} \\ r_{t-1} \end{bmatrix} \in \mathbb{R}^2 \tag{1}$$



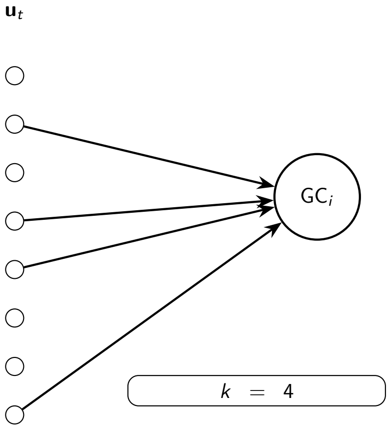
Equation 2

$$u_{t,j} = \begin{cases} \sigma(\beta_j \cdot s_{t,c_j}) & d_j = 0 \text{ (Tonic)} \\ \sigma(\beta_j \cdot (s_{t,c_j} - s_{t-d_j,c_j})) & d_j > 0 \text{ (Phasic)} \end{cases} \quad (2)$$



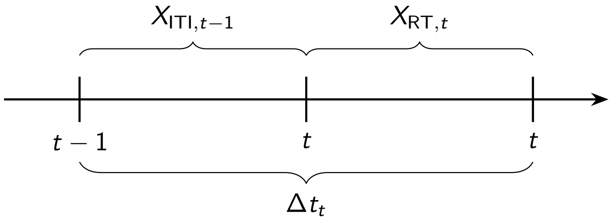
Equation 3

$$\mathbf{W}_{in} \in \mathbb{R}^{N_{GC} \times d_{channels}} \quad \text{(CSR, Row-Degree} = 4) \tag{3}$$



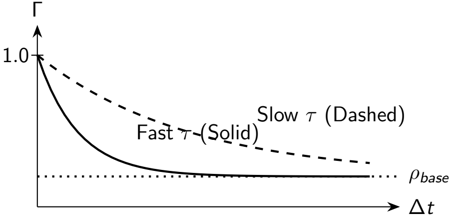
Equation 4

$$\Delta t_t = X_{\mathsf{ITI},t-1} + X_{\mathsf{RT},t} \tag{4}$$



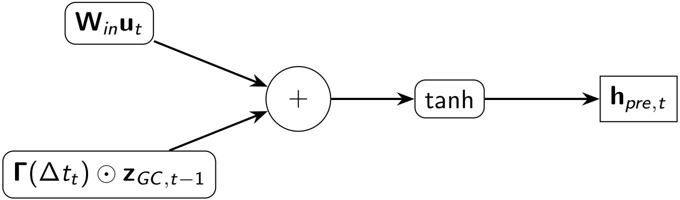
Equation 5

$$\boldsymbol{\Gamma}(\Delta t_t) = \boldsymbol{\rho}_{base} + (\mathbf{1} - \boldsymbol{\rho}_{base}) \odot \exp\left(-\frac{\Delta t_t}{\boldsymbol{\tau}}\right) \qquad (5)$$



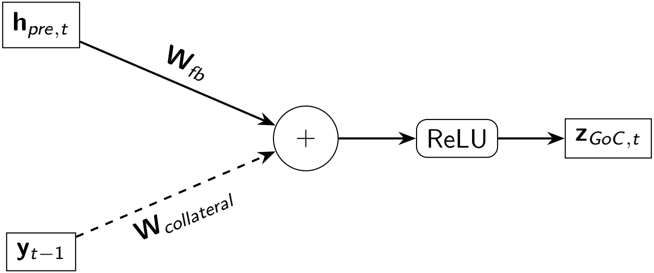
Equation 6

$$\mathbf{h}_{pre,t} = \tanh\left(\mathbf{W}_{in}\mathbf{u}_t + \mathbf{\Gamma}(\Delta t_t) \odot \mathbf{z}_{GC,t-1}\right) \tag{6}$$



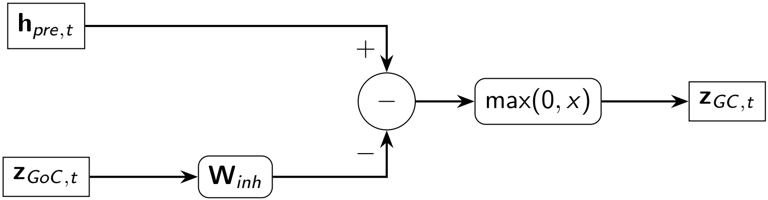
Equation 7

$$\mathbf{z}_{GoC,t} = \mathrm{ReLU}\left(\mathbf{W}_{fb}\mathbf{h}_{pre,t} + \mathbf{W}_{collateral}\mathbf{y}_{t-1}\right) \tag{7}$$



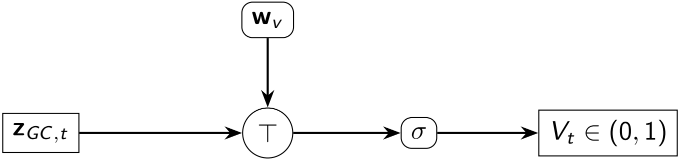
Equation 8

$$\mathbf{z}_{GC,t} = \max(0, \mathbf{h}_{pre,t} - \mathbf{W}_{inh}\mathbf{z}_{GoC,t}) \tag{8}$$



Equation 9

$$V_t = \sigma\left(\mathbf{w}_v^\top \mathbf{z}_{GC,t}\right) \tag{9}$$



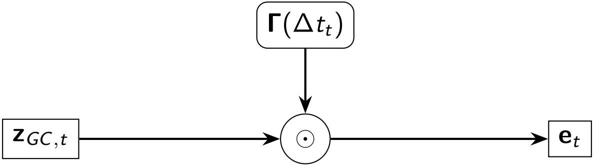
Equation 10

$$\boldsymbol{\pi}_t = \mathrm{Softmax}(\mathbf{W}_\pi \mathbf{z}_{GC,t}) \tag{10}$$

$$\boxed{\mathbf{z}_{GC,t}} \longrightarrow \boxed{\times \mathbf{W}_\pi} \longrightarrow \boxed{\text{Softmax}} \longrightarrow \boxed{\boldsymbol{\pi}_t}$$

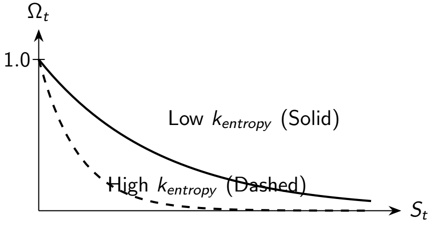
Equation 11

$$\mathbf{e}_t = \boldsymbol{\Gamma}(\Delta t_t) \odot \mathbf{z}_{GC,t} \tag{11}$$



Equation 12

$$S_t = -\sum p_{t,i} \log(p_{t,i}) \quad \Rightarrow \quad \Omega_t = \exp\left(-k_{entropy} \cdot S_t\right) \tag{12}$$



Equation 13

$$\mathbf{w}_{v,t+1} = \max\left(0, \mathbf{w}_{v,t} + \eta \cdot \Omega_t \cdot \delta_t \cdot \mathbf{e}_t\right) \tag{13}$$